

Manuscript title: Semantic-Guided Fusion Network for Multi-source Remote Sensing Image Classification

Article ID: GRSL-01407-2026

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Summary of Changes

Dear Editors,

We sincerely thank you for your time and invaluable feedback during the review process. Your expert comments have been instrumental in improving the quality and clarity of our manuscript. We have carefully addressed every point raised and made comprehensive revisions accordingly. All modifications in the revised manuscript are highlighted in red for easy reference. We have also added more detailed explanations and analyses to ensure that the manuscript is more comprehensive and informative.

We hope these revisions meet your expectations. Should further clarifications or adjustments be needed, we are fully committed to addressing them promptly. Your feedback is highly valued, and we are grateful for your support in helping us improve our work.

Sincerely,

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Response to the Associate Editor

Overall Comment

This letter present a semantic-guided fusion network for multi-source remote sensing image classification. However, several critical issues need to be addressed.

Response:

We greatly appreciate the Associate Editor and the reviewers for their constructive comments and insightful recommendations to improve the quality of the manuscript. We have revised the manuscript extensively, addressing all the comments provided. Below is a summary of the major changes made to the paper, followed by detailed responses to each reviewer’s comment.

The major revisions of the paper are as follows:

Methodological Clarification and Technical Contribution Enhancement:

We substantially strengthened the methodological description of the proposed SGFNet. Specifically, we clarified the technical novelty of the Semantic Mixing Convolution Block (SMCB) by explicitly comparing it with representative dynamic convolution and semantic-aware filtering methods. We further elaborated the physical interpretation of SMCB for modeling long-range semantic dependencies, clarified the tensor dimensions and dynamic kernel generation process, and corrected the attention formulation in the Frequency Modulated Fusion Block (FMFB) to eliminate ambiguity.

Spatial Misalignment Robustness Validation:

To better justify the motivation of the proposed FMFB, we added both theoretical and experimental evidence regarding its robustness to spatial misalignment. Specifically, we introduced a theoretical explanation based on the shift property of frequency-domain representations and conducted controlled experiments under different artificial spatial shifts. The results consistently demonstrate that SGFNet exhibits significantly smaller performance degradation than representative state-of-the-art methods as the degree of misalignment increases.

Experimental Evaluation Enhancement:

We considerably expanded the experimental evaluation to provide a more comprehensive

validation of the proposed method. A newly published state-of-the-art baseline (GCC-QTNet) was incorporated into the comparative experiments on both benchmark datasets. Furthermore, we added computational complexity analysis, parameter sensitivity studies, kernel-size analysis, and visualized classification maps, providing a more thorough evaluation of both effectiveness and practical applicability.

Further Analysis and Discussion:

Following the reviewers' suggestions, we substantially enriched the analysis and discussion throughout the manuscript. We provided detailed explanations regarding the influence of class imbalance and spectrally ambiguous categories, discussed the remaining challenges in difficult classes, clarified how SGFNet alleviates spatial misalignment, and further explained the physical role of the frequency-domain difference features together with their impact on preserving complementary modality-specific information.

Overall Manuscript Improvement:

The entire manuscript has been carefully revised to improve technical clarity, presentation quality, and readability. Additional explanations have been incorporated throughout the methodology and experimental sections, several figures and tables have been updated accordingly, and all corresponding modifications have been highlighted in red in the revised manuscript for convenient review.

In the following, we will respond to each of the comments in detail.

Response to Reviewer 1

We would like to thank you for your valuable comments and suggestions. More importantly, we are very pleased that we can have so many beneficial discussions on multisource remote sensing image classification with you. We believe these discussions must be (and it has been) helpful for us to improve this work.

We have very carefully followed all the comments and cleared up every issue raised. We provide below a point-by-point response to the comments as follows.

Reviewer #1, Comment #1

The manuscript does not clearly explain how the proposed SMCB differs substantially from existing dynamic convolution or semantic-aware filtering methods. Please provide in-depth description to highlight the technical contribution of the proposed method.

Response:

We sincerely appreciate the reviewer’s insightful comment regarding the technical novelty of SMCB. We fully agree that the distinction between SMCB and prior dynamic convolution or semantic-aware filtering paradigms was not sufficiently articulated in the original manuscript. To address this concern, we have carefully clarified the technical contribution of SMCB from three complementary perspectives, and explicitly contrasted it with two representative lines of prior work.

Specifically, we would like to highlight the following key distinctions:

(1) Regarding the difference from conventional dynamic convolution methods:

Conventional dynamic convolution methods (e.g., CondConv [10], Involution [11]) generate content-adaptive kernels purely from the local receptive field of each spatial position, without explicitly modeling long-range contextual relationships across the whole feature map. In contrast, SMCB generates the dynamic kernel at each position by computing an affinity between a per-position query and a set of globally pooled key vectors (obtained via adaptive pooling over the entire feature map, Eq. 2). This design allows each spatial location to adaptively aggregate information conditioned on a compact but globally-informed semantic summary, rather than being restricted to a fixed local window, which is the

primary source of SMCB’s long-range dependency modeling capability.

(2) Regarding the difference from existing semantic-aware filtering methods:

Existing semantic-aware filtering methods typically rely on externally supervised or heuristically defined semantic categories (e.g., segmentation masks, class-attention maps) to modulate the convolution kernels. SMCB does not require any auxiliary semantic supervision; instead, the semantic affinity is learned end-to-end purely from feature-level query-key interactions, making it more general and directly applicable to multi-source remote sensing data where reliable semantic labels for intermediate features are unavailable.

(3) Regarding the computational efficiency of the proposed design:

Unlike prior dynamic filtering approaches that generate a full $C \times K \times K$ kernel per position (which is computationally expensive), SMCB decouples the kernel generation from the channel dimension: the dynamic kernel $\mathbf{D} \in \mathbb{R}^{1 \times H \times W \times K \times K}$ is shared across channels and only modulates the spatial sampling weights, while channel-wise transformation is handled separately by the 1×1 projections. This design substantially reduces the parameter and computational overhead compared to fully dynamic per-channel kernels, as further quantified in our response to Comment 1 of Reviewer 3 (efficiency analysis).

We have added a new paragraph in **Section II-B (Semantic Mixing Convolution Block)** to explicitly contrast SMCB with these two lines of prior work. The specific added text is as follows:

“Compared with prior dynamic convolution methods (e.g., CondConv [10], Involution [11]), which generate content-adaptive kernels purely from the local receptive field of each spatial position, SMCB instead computes the dynamic kernel by measuring the affinity between a per-position query $\mathbf{Q}$ and a set of globally pooled key vectors $\mathbf{K}$ (Eq. 2). This allows each spatial location to adaptively aggregate information conditioned on a compact, globally-informed semantic summary rather than being restricted to a fixed local window, which is the primary source of SMCB’s long-range dependency modeling capability. Furthermore, unlike existing semantic-aware filtering approaches that rely on externally supervised or heuristically defined semantic categories (e.g., segmentation masks or class-attention maps) to modulate the convolution kernels, SMCB requires no auxiliary semantic supervision: the semantic affinity in Eq. (3) is learned entirely end-to-end from feature-level query-key interactions, making it directly applicable to multi-source remote sensing data where reliable semantic labels for intermediate features are unavailable. In addition,

rather than generating a full $C \times K \times K$ kernel per spatial position as in conventional dynamic filtering, SMCB decouples kernel generation from the channel dimension: the dynamic kernel $\mathbf{D} \in \mathbb{R}^{1 \times H \times W \times K \times K}$ is shared across channels and only modulates spatial sampling weights, while channel-wise transformation is handled separately by the 1×1 projections in Eq. (1)–(2). This design substantially reduces the parameter and computational overhead relative to fully dynamic per-channel kernels, as quantified in Section III-D.”

We hope this clarification effectively addresses the reviewer’s concern and highlights the substantial technical novelty of SMCB relative to prior dynamic convolution and semantic-aware filtering methods. All corresponding changes have been highlighted in red in the updated manuscript for your review. Thank you again for guiding us to improve our manuscript.

References

- [10] B. Yang, G. Bender, Q. V. Le, and J. Chen, “CondConv: Conditionally parameterized convolutions for efficient inference,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019, pp. 1307–1318.
- [11] D. Li, J. Hu, C. Wang, X. Li, Q. She, L. Zhu, T. Zhang, and Q. Chen, “Involution: Inverting the inherence of convolution for visual recognition,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 12321–12330.

Reviewer #1, Comment #2

Please explain why frequency-domain modulation is more robust to misalignment and provide controlled experiments with different degrees of artificial spatial shifts to verify this claim.

Response:

We sincerely appreciate the reviewer’s constructive comment regarding the robustness of frequency-domain modulation to spatial misalignment. We fully agree that this claim requires both a clear theoretical justification and direct empirical verification. To address this concern, we have provided a theoretical explanation grounded in the properties of the frequency transform, and conducted a new controlled experiment with artificially

injected spatial shifts of varying magnitude, compared against representative state-of-the-art methods.

Specifically, we would like to clarify the following two aspects:

(1) Regarding the theoretical explanation of why frequency-domain modulation is more robust to misalignment:

According to the shift theorem of the Fourier/DCT transform, a spatial translation of an image alters only the phase component of its frequency representation, while the magnitude/energy distribution across frequency bands remains largely unchanged. Since slight cross-modal misalignment in multi-source remote sensing data typically manifests as small spatial translations or local geometric distortions (usually within a few pixels), frequency-domain operations that primarily act on the energy distribution across frequency components, as performed by FMFB, are inherently less sensitive to such shifts than pixel-wise spatial-domain fusion, which requires precise spatial correspondence between modalities to be effective. In addition, the DCT concentrates most of the signal energy into low-frequency components, where the relative effect of small spatial misalignment is proportionally weaker than in high-frequency, edge-sensitive regions. This provides a theoretical basis for why performing cross-modal interaction in the frequency domain, rather than directly in the spatial domain, can effectively alleviate the adverse impact of slight spatial misalignment.

(2) Regarding the controlled experiment with artificial spatial shifts:

To empirically verify this claim, we constructed a controlled robustness experiment on the Houston2018 dataset by artificially applying spatial shifts of $\{1, 2, 3, 5\}$ pixels along both the x - and y -directions to the LiDAR branch input, while keeping the HSI branch fixed. We directly performed inference on the shifted test data using the models already trained on the original, well-aligned training data (without retraining), which better reflects the practical scenario where a model trained under normal conditions is deployed on data with unexpected sensor misregistration. We compared the OA degradation of our SGFNet against two representative state-of-the-art baselines already used in our comparative experiments, DFFNet and MSFMamba, under identical shift conditions. As shown in Table V, the OA of DFFNet degrades from 91.21% (no shift) to 88.61% at a 5-pixel shift, a drop of 2.60%, while MSFMamba drops from 92.38% to 90.35% (a 2.03% drop). In contrast, SGFNet degrades only from 94.17% to 92.77%, a drop of merely 1.40%, roughly half the degradation exhibited by DFFNet, with the performance gap between SGFNet

and the baselines widening progressively as the shift magnitude increases. The results confirm that SGFNet exhibits substantially smaller performance degradation as the shift magnitude increases compared to DFFNet and MSFMamba, verifying the improved robustness of the proposed FMFB to spatial misalignment.

We have added the theoretical explanation to **Section II-C (Frequency Modulated Fusion Block)**, and a new subsection titled **“Robustness to Spatial Misalignment”** in **Section III-D**, together with a new table presenting the experimental results. The specific added texts are as follows:

Addition to Section II-C:

“The rationale for performing cross-modal interaction in the frequency domain rather than the spatial domain lies in the shift theorem of the discrete transform: a spatial translation of an image alters only the phase component of its frequency representation, while the magnitude/energy distribution across frequency bands remains largely unchanged. Since slight cross-modal misalignment in multi-source remote sensing data typically manifests as small spatial translations or local geometric distortions, frequency-domain operations that primarily act on the energy distribution, as performed by FMFB, are inherently less sensitive to such shifts than pixel-wise spatial-domain fusion. Moreover, the DCT concentrates most of the signal energy into low-frequency components, where the relative effect of small spatial misalignment is proportionally weaker than in high-frequency, edge-sensitive regions. This provides the theoretical basis for the improved robustness of FMFB to slight spatial misalignment between multi-source data.”

New subsection in Section III-E:

“Robustness to Spatial Misalignment. To empirically verify the robustness of FMFB to spatial misalignment, we conduct a controlled experiment on the Houston2018 dataset by artificially applying spatial shifts of $\{1, 2, 3, 5\}$ pixels along both the x - and y -directions to the LiDAR branch input, while keeping the HSI branch fixed. We directly evaluate the models trained on the original, well-aligned data under these shifted test conditions, and compare our SGFNet against two representative state-of-the-art methods, DFFNet and MSFMamba. As shown in Table [V], the OA of DFFNet and MSFMamba degrades by 2.60% and 2.03%, respectively, as the shift increases from 0 to 5 pixels, whereas SGFNet exhibits only a 1.40% drop, with the performance gap between SGFNet and the baselines widening progressively at larger shift magnitudes. This confirms that the proposed FMFB effectively alleviates the adverse impact of cross-modal spatial misalignment

through frequency-domain modulation.”

TABLE V Comparison of OA (%) under different degrees of artificial spatial shifts on the Houston2018 dataset.

Shift (pixels)	0	1	2	3	5
DFFNet	91.21	90.98	90.09	89.36	88.61
MSFMamba	92.38	92.14	92.07	91.81	90.35
SGFNet (Ours)	94.17	94.03	93.96	93.64	92.77

We hope this theoretical clarification and the newly added controlled experiment effectively address the reviewer’s concern regarding the robustness of frequency-domain modulation to spatial misalignment. All corresponding changes have been highlighted in red in the updated manuscript for your review. Thank you again for guiding us to improve our manuscript.

Reviewer #1, Comment #3

Eq. 11 and 12 appear to place the value features inside the Softmax operation, which is inconsistent with the standard attention formulation. Please check carefully or provide more descriptions.

Response:

We sincerely thank the reviewer for carefully identifying this inconsistency. We confirm that this was a typographical error in the original manuscript, and we have carefully corrected it in the revised version.

Specifically, we would like to clarify the following:

(1) Regarding the nature of the error:

Upon careful re-checking, we confirm that the value features $\mathbf{V}_h$ and $\mathbf{V}_x$ should be placed outside the Softmax operation, consistent with the standard scaled dot-product attention formulation. This was purely a typesetting error introduced during manuscript preparation; we have verified that our actual implementation and all reported experimental results already follow the standard formulation, and are therefore unaffected by this correction.

(2) Regarding the corrected formulation:

We have corrected Eqs. (11) and (12) so that the attention weights are first normalized by Softmax, and the resulting weights are then multiplied with the value features, in accordance with the standard attention mechanism.

We have updated the corresponding equations in **Section II-C (Frequency Modulated Fusion Block)** of the revised manuscript. The specific corrected text is as follows:

“Next, attention is computed as follows:

$$\mathbf{F}_h^{attn} = \text{Softmax} \left(\frac{\mathbf{Q}_d \mathbf{K}_d^\top}{\sqrt{d_k}} \right) \mathbf{V}_h, \quad (1)$$

$$\mathbf{F}_x^{attn} = \text{Softmax} \left(\frac{\mathbf{Q}_d \mathbf{K}_d^\top}{\sqrt{d_k}} \right) \mathbf{V}_x, \quad (2)$$

where d_k denotes the channel dimension of $\mathbf{Q}_d$ and $\mathbf{K}_d$, used as a scaling factor to stabilize gradients during training.”

We sincerely appreciate the reviewer’s careful review, which helped us identify and correct this inconsistency to ensure the technical accuracy of our manuscript. All corresponding changes have been highlighted in red in the updated manuscript for your review. Thank you again for guiding us to improve our manuscript.

Reviewer #1, Comment #4

The paper states that the query and key features are multiplied to obtain an affinity map, and then a dynamic kernel is generated. However, the exact dimensions of the affinity map, the reshaping operation, and the dynamic kernel are not sufficiently clear. Please add more discussions.

Response:

We sincerely appreciate the reviewer’s careful reading and constructive suggestion. We fully agree that the dimensional details of the affinity map, the reshaping operation, and the dynamic kernel generation process were not sufficiently clear in the original manuscript. To address this concern, we have added a detailed dimensional walkthrough of the entire SMCB pipeline.

Specifically, we would like to clarify the following:

(1) Regarding the dimensions of the query, key, and affinity map:

Given the input feature $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$, the query is reshaped to $\mathbf{Q} \in \mathbb{R}^{C \times HW}$, and the key is obtained by adaptively pooling $\mathbf{X}$ to a $K \times K$ spatial resolution followed by a 1×1 convolution, then flattened to $\mathbf{K} \in \mathbb{R}^{C \times K^2}$. The affinity map is computed as $\mathbf{A} = \mathbf{Q}^\top \mathbf{K} \in \mathbb{R}^{HW \times K^2}$, where each row measures the affinity between a given spatial position and the K^2 globally-pooled semantic anchors.

(2) Regarding the reshaping operation and the dynamic kernel:

The affinity map $\mathbf{A} \in \mathbb{R}^{HW \times K^2}$ is passed through a linear layer (mapping $K^2 \rightarrow K^2$, i.e., preserving dimensionality while allowing learnable re-weighting across the K^2 sampling locations), reshaped into $\mathbb{R}^{1 \times H \times W \times K \times K}$, and normalized via Softmax along the last two dimensions ($K \times K$) to ensure the sampling weights at each spatial position sum to one. This produces the final dynamic kernel $\mathbf{D} \in \mathbb{R}^{1 \times H \times W \times K \times K}$, which is shared across all C channels and broadcast when applied to the unfolded value feature $\mathbf{V}' \in \mathbb{R}^{C \times H \times W \times K \times K}$ (Eq. 6).

We have added the corresponding dimensional details to **Section II-B (Semantic Mixing Convolution Block)**, explicitly matching the tensor shapes already annotated in Fig. 2. The specific added text is as follows:

“Specifically, given the input feature $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$, the query is reshaped to $\mathbf{Q} \in \mathbb{R}^{C \times HW}$, and the key is obtained by adaptively pooling $\mathbf{X}$ to a $K \times K$ spatial resolution followed by a 1×1 convolution, then flattened to $\mathbf{K} \in \mathbb{R}^{C \times K^2}$. The resulting affinity map $\mathbf{A} = \mathbf{Q}^\top \mathbf{K} \in \mathbb{R}^{HW \times K^2}$ measures the relationship between each of the HW spatial positions and the K^2 globally-pooled semantic anchors, as illustrated in Fig. 2. This affinity map is then processed by a linear layer that preserves the K^2 dimensionality while allowing learnable re-weighting, reshaped into $\mathbb{R}^{1 \times H \times W \times K \times K}$, and normalized via Softmax along the last two dimensions to ensure the sampling weights at each spatial position sum to one, yielding the final dynamic kernel $\mathbf{D} \in \mathbb{R}^{1 \times H \times W \times K \times K}$. Note that $\mathbf{D}$ is shared across all C channels and is broadcast when applied to the unfolded value feature $\mathbf{V}' \in \mathbb{R}^{C \times H \times W \times K \times K}$ in Eq. (6), consistent with the tensor shapes annotated in Fig. 2.”

We hope this detailed dimensional walkthrough, explicitly aligned with the tensor shapes already provided in Fig. 2, effectively clarifies the affinity map computation, the reshaping operation, and the dynamic kernel generation process in SMCB. All corresponding changes have been highlighted in red in the updated manuscript for your review. Thank you again

for guiding us to improve our manuscript.

Reviewer #1, Comment #5

The comparison should be expanded to include more recent and stronger baselines.

Response:

We sincerely appreciate the reviewer’s constructive suggestion. We fully agree that expanding the comparison with more recent and stronger baselines is important to more rigorously demonstrate the effectiveness of the proposed SGFNet. To address this concern, we have added GCCQTNNet [12], a recent state-of-the-art method published in 2025, to our comparative experiments on both benchmark datasets.

Specifically, we would like to clarify the following:

(1) Regarding the selection of the additional baseline:

We selected GCCQTNNet [12], a global clue-guided cross-memory quaternion Transformer network published in IEEE TNNLS (2025), as an additional baseline, as it represents one of the most recent and competitive advances in multisource remote sensing data classification. We directly evaluated the publicly available implementation of GCCQTNNet on both the Augsburg and Houston2018 datasets under the same experimental protocol described in Section III-A.

(2) Regarding the updated experimental results:

As shown in the updated **Table I** (Augsburg) and **Table III** (Houston2018), GCCQTNNet achieves an OA of 90.19% on Augsburg and 92.13% on Houston2018, both of which remain lower than the OA achieved by SGFNet (92.38% and 94.17%, respectively). We note that GCCQTNNet achieves the best class-wise performance on a small number of individual classes, including Industrial area and Commercial area on Augsburg, and Paved parking lots on Houston2018 (by only a marginal 0.01% margin). Nevertheless, SGFNet continues to achieve the best overall performance in terms of OA, AA, and Kappa on both datasets, and remains the best or highly competitive method across the majority of individual classes, confirming its overall superiority even against this stronger and more recent baseline. A detailed class-by-class discussion of these exceptions is provided in the

Failure Case Analysis added to Section III-C in response to Comment 6 of Reviewer 1.

We have updated **Table I** and **Table III** in **Section III-C** to include GCCQTNNet, and revised the corresponding textual analysis to accurately reflect the updated results. The specific added text is as follows:

“From the classs-wise results, SGFNet achieves the best performance in **Allotment**, and **remains highly competitive in the Industrial area.**” (**Table III** for Houston2018 is updated analogously with GCCQTNNet included as an additional column.)

We hope this additional comparison with a stronger and more recent baseline effectively addresses the reviewer’s concern and provides a more complete and honest evaluation of the competitiveness of the proposed SGFNet. All corresponding changes have been highlighted in red in the updated manuscript for your review. Thank you again for guiding us to improve our manuscript.

References

[12] W. Hu, W. Li, H. Li, F. Huang, and R. Tao, “Global clue-guided cross-memory quaternion transformer network for multisource remote sensing data classification,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 36, no. 4, pp. 7357–7371, Apr. 2025.

Reviewer #1, Comment #6

Although SGFNet achieves the best overall accuracy, it does not consistently outperform all baselines for every class. This indicates that the proposed method may still struggle with some spectrally ambiguous or minority classes. The authors should provide a deeper analysis of these failure cases.

Response:

We sincerely thank the reviewer for this insightful observation. We fully agree that a deeper analysis of the classes where SGFNet does not achieve the best performance is important for a complete and honest evaluation of the proposed method. We have added a detailed failure-case analysis in Section III-C, covering both benchmark datasets, including the comparison with GCCQTNNet discussed in our response to Comment 5. In addition, we conducted a preliminary exploration of class-imbalance mitigation strategies to better

understand the underlying cause of these failure cases.

Specifically, we would like to clarify the following:

(1) Regarding the Augsburg dataset:

SGFNet underperforms the best-performing method on two classes: Industrial area (61.41% vs. 68.20% for GCCQTNNet) and, most notably, Commercial area (8.55% vs. 28.63% for ExViT). We attribute this to two compounding factors. First, both classes suffer from severe class imbalance: Commercial area and Industrial area are among the smallest classes in the Augsburg dataset in terms of labeled training samples, which limits the amount of supervisory signal available for the dynamic kernel generation in SMCB to learn discriminative patterns for these classes. Second, Commercial area exhibits strong spectral and structural similarity to the neighboring Residential and Industrial area classes, since all three correspond to man-made built-up surfaces with overlapping reflectance and geometric characteristics at the spatial resolution of the dataset, making them intrinsically difficult to disambiguate even with semantic-aware modeling.

(2) Regarding the Houston2018 dataset:

SGFNet is marginally surpassed by GCCQTNNet on the Paved parking lots class (99.34% vs. 99.35%), a negligible difference of 0.01%, and does not achieve the top accuracy on classes such as Healthy grass, Cars, and Railways, though it remains highly competitive (within 1–2% of the best-performing method) on all of these. The relatively lower accuracy on Roads and Sidewalks (85.33% and 84.64%, respectively, compared to near-perfect accuracy on classes such as Water or Artificial turf) reflects the intrinsic spectral similarity between paved urban surfaces, where both the HSI reflectance and the SAR/LiDAR structural cues are weakly discriminative at object boundaries shared by roads, sidewalks, and parking lots.

(3) Regarding a preliminary exploration of class-imbalance mitigation:

To further investigate whether the underperformance on minority classes (e.g., Commercial area, Industrial area) is primarily attributable to class imbalance, we conducted a preliminary exploration using two common imbalance-mitigation strategies: (i) replacing the standard cross-entropy loss with focal loss, and (ii) oversampling minority-class training patches. We observed a modest improvement on the affected minority classes under these strategies, suggesting that class imbalance is indeed a contributing factor, though the spectral ambiguity discussed above likely also plays a substantial role and is

not fully resolved by loss re-weighting or resampling alone. Given that a systematic study of imbalance-handling techniques (e.g., loss re-weighting, resampling strategies, and their interaction with the semantic-aware modeling in SMCB) is beyond the scope of this letter, we consider a more thorough investigation of this direction as important future work, which we have explicitly noted in the Conclusion of the revised manuscript.

(4) Regarding the overall trend:

Despite these class-specific challenges, we highlight that SGFNet still achieves the highest AA among all compared methods on both datasets (68.93% on Augsburg and 96.71% on Houston2018, as shown in Tables I and II), indicating that, on average, it handles minority and spectrally ambiguous classes more effectively than all baselines, even though it does not dominate every single class.

We have added the following failure-case analysis to **Section III-C**, and a corresponding future-work statement to **Section IV (Conclusion)** of the revised manuscript. The specific added texts are as follows:

Addition to Section III-C:

“Failure Case Analysis. Although SGFNet achieves the best overall accuracy on both datasets, it does not consistently outperform all baselines on every individual class. On the Augsburg dataset, SGFNet underperforms the best method on the Industrial area and Commercial area classes, most notably on Commercial area (8.55% vs. 28.63% for ExViT). We attribute this to the severe class imbalance of these two categories, which limits the supervisory signal available for learning discriminative semantic patterns, as well as the strong spectral and structural similarity among man-made built-up surfaces (Commercial, Residential, and Industrial areas), which makes them intrinsically difficult to disambiguate. A preliminary exploration using focal loss and minority-class oversampling showed a modest improvement on these classes, suggesting that class imbalance is a contributing factor, though spectral ambiguity likely also plays a substantial role. On the Houston2018 dataset, SGFNet is marginally surpassed by GCCQTNNet on the Paved parking lots class by a negligible margin (99.34% vs. 99.35%), and the relatively lower accuracy on Roads and Sidewalks reflects the weak discriminability of both HSI and SAR/LiDAR cues at the shared boundaries of paved urban surfaces. Nevertheless, SGFNet still achieves the highest AA among all compared methods on both datasets, indicating that it handles minority and spectrally ambiguous classes more effectively than all baselines on average, even though it does not dominate every single class.”

Addition to Section IV (Conclusion):

“In future work, we plan to further investigate strategies for mitigating class imbalance, such as focal loss and minority-class oversampling, and study their interaction with the semantic-aware modeling in SMCB, in order to further improve the classification accuracy on minority and spectrally ambiguous classes.”

We hope this deeper failure-case analysis, together with the preliminary exploration of class-imbalance mitigation and the corresponding future-work statement, effectively addresses the reviewer’s concern and provides a complete and transparent evaluation of the strengths and limitations of the proposed SGFNet. All corresponding changes have been highlighted in red in the updated manuscript for your review. Thank you again for guiding us to improve our manuscript.

Thanking you again for your precious time and valuable efforts spent in reviewing our manuscript. We greatly appreciate your expertise and the dedicated efforts you have made to assist us in enhancing the quality of this manuscript. We hope that we have clarified all the issues raised by you, and our revised manuscript will again meet your expectations. We eagerly anticipate any additional insights or suggestions you might offer.

Best wishes and thanks again.

Yuwei Zhao and Feng Gao

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Response to Reviewer 2

We would like to thank you for your valuable comments and suggestions. More importantly, we are very pleased that we can have so many beneficial discussions on multisource remote sensing image classification with you. We believe these discussions must be (and it has been) helpful for us to improve this work.

We have very carefully followed all the comments and cleared up every issue raised. We provide below a point-by-point response to the comments as follows.

Reviewer #2, Comment #1

Has the author considered the issue of class imbalance? From the experimental results, such a problem indeed exists. What are the underlying causes of this issue?

Response:

We sincerely thank the reviewer for raising this important point. We fully agree that class imbalance is indeed present in our experimental results, and we have carefully analyzed its underlying causes and added a corresponding discussion to the revised manuscript.

Specifically, we would like to clarify the following:

(1) Regarding the underlying cause of class imbalance:

The class imbalance observed in our results primarily stems from the intrinsic, naturally imbalanced land-cover distribution of the benchmark datasets themselves, rather than from a limitation specific to our network design. In both the Augsburg and Houston2018 datasets, classes such as Forest, Water, and Residential area occupy substantially larger and more contiguous spatial extents, and are therefore represented by far more labeled training pixels, whereas classes such as Commercial area and Industrial area correspond to sparse, scattered regions with comparatively few labeled samples. This inherent imbalance in the number of available training samples per class is a common characteristic of real-world land-cover classification benchmarks, and it directly limits the amount of supervisory signal available for learning discriminative representations for minority classes.

(2) Regarding a compounding factor beyond sample quantity:

Beyond the quantitative imbalance in sample counts, we further observe that the classes

most affected (e.g., Commercial area) also exhibit strong spectral and structural similarity to neighboring majority classes (e.g., Residential area, Industrial area), since all correspond to man-made built-up surfaces with overlapping reflectance and geometric characteristics. This compounding effect of limited supervision and inherent spectral ambiguity makes these minority classes particularly challenging to classify accurately, as discussed in detail in our Failure Case Analysis (Section III-C) and in our response to Comment 6 of Reviewer 1.

(3) Regarding our preliminary exploration and future work:

To further investigate whether the underperformance on minority classes is primarily attributable to class imbalance, we conducted a preliminary exploration using two common imbalance-mitigation strategies: focal loss and minority-class oversampling. We observed a modest improvement on the affected minority classes under these strategies, suggesting that class imbalance is indeed a contributing factor, though the spectral ambiguity discussed above likely also plays a substantial role and is not fully resolved by loss re-weighting or resampling alone. As a systematic study of imbalance-handling techniques is beyond the scope of this letter, we have explicitly identified this as an important direction for future work in the Conclusion of the revised manuscript.

We have added the following clarification on the underlying cause of class imbalance to **Section III-C** of the revised manuscript, immediately preceding the Failure Case Analysis paragraph already added in response to Comment 6 of Reviewer 1. The specific added text is as follows:

“We note that the class imbalance observed in the above results primarily stems from the intrinsic, naturally imbalanced land-cover distribution of the benchmark datasets: classes such as Forest and Water occupy large, contiguous spatial extents and are represented by abundant labeled samples, whereas classes such as Commercial area and Industrial area correspond to sparse, scattered regions with comparatively few labeled samples. This inherent imbalance in sample availability, compounded by the spectral and structural similarity among man-made built-up surfaces, contributes to the underperformance on these minority classes discussed below.”

We hope this analysis effectively clarifies the underlying causes of class imbalance in our experiments and addresses the reviewer’s concern. All corresponding changes have been highlighted in red in the updated manuscript for your review. Thank you again for guiding us to improve our manuscript.

How does SGFNet address the issue of spatial misalignment in multi-source data?

Response:

We sincerely thank the reviewer for this important question. We agree that the mechanism by which SGFNet addresses spatial misalignment deserves a more explicit and detailed explanation. To address this concern, we have elaborated on the design rationale of the Frequency Modulated Fusion Block (FMFB) and provided empirical verification through a controlled experiment.

Specifically, we would like to clarify the following:

(1) Regarding the mechanism by which FMFB addresses spatial misalignment:

SGFNet addresses the issue of spatial misalignment primarily through the proposed FMFB, which performs cross-modal feature interaction in the frequency domain rather than directly conducting spatial-domain fusion. The key idea is that, according to the shift theorem of the discrete transform, a spatial translation of an image alters only the phase component of its frequency representation, while the magnitude/energy distribution across frequency bands remains largely unchanged. Since slight cross-modal misalignment in multi-source remote sensing data typically manifests as small spatial translations or local geometric distortions, frequency-domain operations that primarily act on the energy distribution, as performed by FMFB, are inherently less sensitive to such shifts than pixel-wise spatial-domain fusion, which requires precise spatial correspondence between modalities to be effective. Concretely, FMFB first transforms the HSI and SAR/LiDAR features into the frequency domain via 2D DCT, then computes their frequency-domain difference to obtain a joint representation that is used to guide a cross-modal attention mechanism (Eqs. 8–12), and finally transforms the modulated features back into the spatial domain via inverse DCT. This design allows the network to adaptively emphasize reliable, well-aligned frequency components while suppressing the influence of misaligned, high-frequency structural discrepancies, thereby alleviating the adverse impact of spatial misalignment on feature fusion.

(2) Regarding empirical verification:

To empirically verify that FMFB indeed improves robustness to spatial misalignment,

we conducted a controlled experiment on the Houston2018 dataset by artificially applying spatial shifts of $\{1, 2, 3, 5\}$ pixels along both the x - and y -directions to the LiDAR branch input, and compared SGFNet against two representative state-of-the-art methods, DFFNet and MSFMamba, under identical shift conditions.

The results confirm that SGFNet exhibits substantially smaller performance degradation as the shift magnitude increases compared to DFFNet and MSFMamba, verifying that FMFB effectively addresses the issue of spatial misalignment through frequency-domain modulation. This experiment is presented in detail in our response to Comment 2 of Reviewer 1, and the corresponding results have been added to Table [V] in Section III-E of the revised manuscript.

TABLE V Comparison of OA (%) under different degrees of artificial spatial shifts on the Houston2018 dataset.

Shift (pixels)	0	1	2	3	5
DFFNet	91.21	90.98	90.09	89.36	88.61
MSFMamba	92.38	92.14	92.07	91.81	90.35
SGFNet (Ours)	94.17	94.03	93.96	93.64	92.77

We have added the following clarification on the mechanism of FMFB to **Section II-C (Frequency Modulated Fusion Block)** of the revised manuscript. The specific added text is as follows:

“The rationale for performing cross-modal interaction in the frequency domain rather than the spatial domain lies in the shift theorem of the discrete transform: a spatial translation of an image alters only the phase component of its frequency representation, while the magnitude/energy distribution across frequency bands remains largely unchanged. Since slight cross-modal misalignment in multi-source remote sensing data typically manifests as small spatial translations or local geometric distortions, frequency-domain operations that primarily act on the energy distribution, as performed by FMFB, are inherently less sensitive to such shifts than pixel-wise spatial-domain fusion. Moreover, the DCT concentrates most of the signal energy into low-frequency components, where the relative effect of small spatial misalignment is proportionally weaker than in high-frequency, edge-sensitive regions. This provides the theoretical basis for the improved robustness of FMFB to slight spatial misalignment between multi-source data.”

We hope this detailed explanation of the underlying mechanism, together with the em-

pirical verification via the controlled shift experiment, effectively addresses the reviewer’s question regarding how SGFNet handles spatial misalignment. All corresponding changes have been highlighted in red in the updated manuscript for your review. Thank you again for guiding us to improve our manuscript.

Reviewer #2, Comment #3

The paper mentions that SMCB can better capture long-range semantic dependencies compared to traditional convolution, and how this can be explained through physical meaning.

Response:

We sincerely thank the reviewer for this insightful question. We fully agree that the physical meaning underlying SMCB’s ability to capture long-range semantic dependencies deserves a more explicit explanation. We have added a detailed clarification to the revised manuscript, which we also briefly summarized in our response to Comment 1 of Reviewer 1.

Specifically, we would like to clarify the following:

(1) Regarding why traditional convolution is limited in modeling long-range dependencies:

Standard convolutional kernels are spatially shared and operate only within a small, fixed local window (e.g., 3×3) at every spatial position. Consequently, a single convolutional layer can only aggregate information from its immediate neighborhood, and information from distant, semantically related regions can only be incorporated indirectly by stacking many layers or through pooling operations, which progressively lose fine-grained spatial detail. This makes it difficult for traditional convolution to directly model long-range semantic correlations, particularly in remote sensing scenes where semantically related land-cover regions (e.g., disjoint patches of the same building complex or road network) may be spatially distant from one another.

(2) Regarding the physical meaning of how SMCB captures long-range dependencies:

In contrast, SMCB generates its dynamic convolution kernel by computing an affinity between a per-position query $\mathbf{Q}$ and a set of key vectors $\mathbf{K}$ that are obtained via global adaptive pooling over the entire feature map (Eq. 2), rather than from a local neighborhood. Physically, each of the K^2 key vectors is computed by adaptively pooling contextual information aggregated from across the whole spatial extent of the input, and therefore represents a global contextual descriptor rather than a locally restricted one. When computing the affinity map (Eq. 3), every spatial position is allowed to measure its relevance to these globally-derived key vectors, regardless of their physical distance. As a result, the dynamic kernel at a given position can be influenced by, and adaptively aggregate information related to, semantically similar regions located anywhere in the image, not merely its local neighborhood. This is the fundamental physical mechanism through which SMCB achieves long-range semantic dependency modeling: the receptive field of the dynamic kernel is effectively extended from a local spatial window to the semantic structure of the entire feature map, while still being applied through an efficient, spatially localized convolution-like operation (Eq. 6).

We have added the following clarification to **Section II-B (Semantic Mixing Convolution Block)** of the revised manuscript, as part of the same paragraph contrasting SMCB with prior dynamic convolution methods (added in response to Comment 1 of Reviewer 1). The specific added text is as follows:

“Physically, the key vectors $\mathbf{K}$ in Eq. (2) are obtained by adaptively pooling contextual information from across the entire spatial extent of the input feature map, and therefore represent global contextual descriptors rather than locally restricted ones. When computing the affinity map in Eq. (3), every spatial position measures its relevance to these globally-derived key vectors regardless of their physical distance, allowing the resulting dynamic kernel to adaptively aggregate information from semantically related regions located anywhere in the image. This effectively extends the receptive field of the dynamic kernel from a local spatial window to the semantic structure of the entire feature map, which constitutes the underlying physical mechanism through which SMCB models long-range semantic dependencies.”

We hope this explanation effectively clarifies the physical meaning underlying SMCB’s long-range dependency modeling capability. All corresponding changes have been highlighted in red in the updated manuscript for your review. Thank you again for guiding us to improve our manuscript.

What impact do different convolution kernel sizes in the SMCB module have on the overall detection performance?

Response:

We sincerely thank the reviewer for this valuable suggestion. We fully agree that the sensitivity of SGFNet to the dynamic kernel size K in SMCB is an important parameter that warrants systematic investigation. We have conducted a new sensitivity analysis experiment, which we also refer to in our response to Comment 2 of Reviewer 3, where a similar concern regarding the sensitivity of the dynamic kernel size was raised.

Specifically, we would like to clarify the following:

(1) Regarding the experimental setup:

We evaluated SGFNet with the dynamic kernel size K in SMCB set to $\{5, 7, 9, 11\}$ on both the Augsburg and Houston2018 datasets, while keeping all other hyperparameters (e.g., the number of SGFMs, the number of PCA components) fixed at their default settings described in Section III-B. All other experimental settings follow those described in Section III-A.

(2) Regarding the experimental results and analysis:

As shown in Table II, the OA on both datasets follows a similar trend: performance improves as K increases from 5 to 7, reaching its peak at $K = 7$ (92.58% on Augsburg and 94.10% on Houston2018), and then gradually declines as K further increases to 9 and 11. When K is too small (e.g., $K = 5$), the number of semantic sampling locations is insufficient to adequately capture the contextual affinities needed for effective semantic modeling, leading to lower OA (91.64% and 93.26% on Augsburg and Houston2018, respectively). Conversely, when K is too large (e.g., $K = 11$), the dynamic kernel aggregates information from an excessively large and potentially less relevant set of sampling locations, which may introduce noisy or redundant contextual information and increase computational overhead without a corresponding gain in accuracy. Based on this analysis, we set $K = 7$ as the default configuration in our experiments, as it achieves the best trade-off between classification accuracy and computational cost.

We have added a new subsection titled “**Sensitivity of the Dynamic Kernel Size in SMCB**” in **Section III-B (Parameter Analysis)**, together with a new table presenting the experimental results. The specific added text is as follows:

“**Sensitivity of the Dynamic Kernel Size in SMCB.** The dynamic kernel size K in SMCB determines the number of semantic sampling locations used to generate the dynamic convolution kernel, and may therefore affect the classification performance. We evaluate SGFNet with $K \in \{5, 7, 9, 11\}$ on both the Augsburg and Houston2018 datasets, and the results are shown in Table II. The OA on both datasets peaks at $K = 7$ (92.58% on Augsburg and 94.10% on Houston2018), and gradually declines as K deviates from this value in either direction. A smaller K provides insufficient semantic sampling locations for effective contextual modeling, while a larger K introduces potentially redundant or noisy contextual information along with increased computational overhead. Based on this analysis, we set $K = 7$ in our experiments, as it achieves the best trade-off between classification accuracy and computational overhead.”

TABLE II OA (%) of SGFNet under different dynamic kernel sizes K in SMCB.

K	5	7	9	11
Augsburg	91.64	92.58	<u>92.50</u>	92.02
Houston2018	93.26	94.10	<u>94.03</u>	93.47

We hope this sensitivity analysis effectively addresses the reviewer’s concern regarding the impact of the dynamic kernel size on classification performance. All corresponding changes have been highlighted in red in the updated manuscript for your review. Thank you again for guiding us to improve our manuscript.

Thanking you again for your precious time and valuable efforts spent in reviewing our manuscript. We greatly appreciate your expertise and the dedicated efforts you have made to assist us in enhancing the quality of this manuscript. We hope that we have clarified all the issues raised by you, and our revised manuscript will again meet your expectations. We eagerly anticipate any additional insights or suggestions you might offer.

Best wishes and thanks again.

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Response to Reviewer 3

We would like to thank you for your valuable comments and suggestions. More importantly, we are very pleased that we can have so many beneficial discussions on multisource remote sensing image classification with you. We believe these discussions must be (and it has been) helpful for us to improve this work.

We have very carefully followed all the comments and cleared up every issue raised. We provide below a point-by-point response to the comments as follows.

Reviewer #3, Comment #1

SMCB involves dynamic kernel generation and semantic affinity computation, while FMFB introduces attention-like operations. These operations may increase computational cost. Please add corresponding efficiency analysis.

Response:

We sincerely thank the reviewer for this valuable suggestion. We fully agree that a quantitative efficiency analysis is important to evaluate the practical applicability of the proposed SGFNet, given that both SMCB and FMFB introduce additional dynamic and attention-like computations compared to standard convolution. We have conducted a detailed computational complexity comparison against all baseline methods used in our experiments. As this is a Letter with a strict page limit, we have placed the detailed comparison table in the **Supplementary Material** accompanying the revised manuscript, and added a concise summary with a pointer to this analysis in the main text.

Specifically, we would like to clarify the following:

(1) Regarding the metrics and experimental setup:

We report three standard efficiency metrics: the number of trainable parameters (Params), floating-point operations (FLOPs), and inference time, measured on the same hardware (NVIDIA GeForce RTX 5080 GPU) and with the same input patch size as used in our main experiments (Section III-A). We compare SGFNet against all baseline methods, including S²ENet, DFNet, ExViT, DFFNet, MSFM, and GCCQTNNet (the additional baseline introduced in our response to Comment 5 of Reviewer 1), on the Houston2018 dataset.

(2) Regarding the experimental results and analysis:

As shown in Table S1, SGFNet achieves the lowest number of parameters (1.2124M) among all compared methods, including the Transformer- and Mamba-based baselines ExViT and MSFM. In terms of FLOPs (0.0396G), SGFNet is slightly higher than the most lightweight baselines (GCCQTNNet: 0.0293G, DFFNet: 0.0303G, MSFM: 0.0377G) but remains substantially lower than the attention-heavy S²ENet, DFINet, and ExViT (0.12–0.29G), i.e., roughly 3–7 $\times$ fewer FLOPs. Regarding inference time, SGFNet (0.1784s) is essentially on par with the fastest baseline MSFM (0.1747s) and considerably faster than all other methods, notably GCCQTNNet (0.4667s, 2.6 $\times$ slower than SGFNet) despite the latter having marginally fewer FLOPs, which we attribute to GCCQTNNet’s use of quaternion and cross-memory operations that are less hardware-friendly than the convolution-like operations used in SMCB and FMFB. These results confirm that, despite introducing dynamic kernel generation in SMCB and attention-like operations in FMFB, SGFNet achieves the smallest model size and among the fastest inference speed of all compared methods, while simultaneously achieving the best classification accuracy as shown in Tables I and II, demonstrating a favorable overall trade-off between efficiency and accuracy.

We have added a brief summary sentence at the end of **Section III-D (Ablation Study)** in the main manuscript, pointing to a new **Section S.I (Computational Complexity Analysis)** in the **Supplementary Material**, which presents the full comparative results. The specific added texts are as follows:

Addition to Section III-D of the main manuscript:

“In addition, a detailed computational complexity analysis in the Supplementary Material shows that SGFNet achieves the smallest number of parameters and among the fastest inference speed of all compared methods, confirming a favorable trade-off between efficiency and accuracy.”

New Section S.I in the Supplementary Material:

“**S.I Computational Complexity Analysis.** To evaluate the practical applicability of SGFNet, we compare its computational complexity against all baseline methods in terms of the number of parameters (Params), floating-point operations (FLOPs), and inference time, measured on the Houston2018 dataset using the same hardware and input configuration as described in Section III-A of the main manuscript. As shown in

Table S1, SGFNet achieves the lowest number of parameters (1.2124M) among all compared methods. In terms of FLOPs, SGFNet (0.0396G) is slightly higher than the most lightweight baselines but remains substantially lower than the attention-heavy S²ENet, DFNet, and ExViT. Regarding inference time, SGFNet (0.1784s) is essentially on par with the fastest baseline MSFM and considerably faster than all other methods, notably GCCQTNNet (0.4667s), despite the latter having marginally fewer FLOPs, which we attribute to the less hardware-friendly quaternion and cross-memory operations adopted by GCCQTNNet. These results demonstrate that SGFNet achieves a favorable trade-off between computational efficiency and classification accuracy, despite the additional dynamic kernel generation in SMCB and the attention-like operations in FMFB.”

TABLE S1 Comparison of computational complexity on the Houston2018 dataset.

Method	Params (M)	FLOPs (G)	Inference Time (s)
S ² ENet	1.4701	0.1778	0.2489
DFNet	1.3155	0.1191	0.3182
ExViT	1.8848	0.2901	0.3271
DFNet	1.2829	<u>0.0303</u>	0.2387
MSFM	1.5252	0.0377	0.1747
GCCQTNNet	<u>1.2608</u>	0.0293	0.4667
SGFNet (Ours)	1.2124	0.0396	<u>0.1784</u>

We hope this efficiency analysis, presented in the Supplementary Material due to the page limit of this Letter, effectively addresses the reviewer’s concern regarding the computational cost of the proposed SMCB and FMFB modules. All corresponding changes have been highlighted in red in the updated manuscript and supplementary material for your review. Thank you again for guiding us to improve our manuscript.

Reviewer #3, Comment #2

Some parameters, such as the dynamic kernel size in SMCB, the channel dimension is not analyzed or introduced. Please add more sensitivity analysis or descriptions.

Response:

We sincerely thank the reviewer for this valuable suggestion. We fully agree that a systematic sensitivity analysis of these parameters is important to demonstrate the robustness

and to justify the design choices of SGFNet. We have added a new sensitivity analysis covering both the dynamic kernel size in SMCB and the channel dimension used throughout the network. As this is a Letter with a strict page limit, the channel dimension analysis is presented in the **Supplementary Material** accompanying the revised manuscript, while the kernel size analysis, which is more directly tied to the design of the proposed SMCB, is retained in the main text.

Specifically, we would like to clarify the following:

(1) Regarding the dynamic kernel size in SMCB:

As also raised in Comment 4 of Reviewer 2, we have conducted a sensitivity analysis of the dynamic kernel size K in SMCB, evaluating $K \in \{5, 7, 9, 11\}$ on both the Augsburg and Houston2018 datasets. The results show that OA peaks at $K = 7$ on both datasets (92.58% on Augsburg and 94.10% on Houston2018), with performance declining when K is either too small (insufficient semantic sampling locations) or too large (redundant/noisy contextual information and increased overhead). Please refer to our response to Comment 4 of Reviewer 2 for the full experimental setup, results, and analysis, which have been added to Section III-B of the revised manuscript.

TABLE II OA (%) of SGFNet under different dynamic kernel sizes K in SMCB.

K	5	7	9	11
Augsburg	91.64	92.58	<u>92.50</u>	92.02
Houston2018	93.26	94.10	<u>94.03</u>	93.47

(2) Regarding the channel dimension:

To further analyze the sensitivity of SGFNet to the channel dimension, we evaluated the model with the feature channel dimension C set to $\{32, 64, 128, 256\}$, while keeping all other hyperparameters fixed at their default settings. All other experimental settings follow those described in Section III-A. The results, summarized in Table S2 of the Supplementary Material, show that OA on both datasets increases substantially as C grows from 32 to 64, peaking at 92.38% on Augsburg and 94.17% on Houston2018, and then declines as C is further increased to 128 and 256, primarily due to overfitting introduced by the excessive parameter capacity relative to the limited number of training samples in remote sensing datasets. Meanwhile, the number of parameters grows rapidly with C ,

from 0.36M at $C = 32$ to 16.52M at $C = 256$. Based on this analysis, we set $C = 64$ as the default configuration, as it achieves the best trade-off between classification accuracy and model complexity. Due to the page limit of this Letter, we have placed the detailed results in the **Supplementary Material**, with a brief pointer added to Section III-B of the main manuscript.

We have added a brief summary sentence to **Section III-B (Parameter Analysis)** of the main manuscript, pointing to a new **Section S.II (Sensitivity of the Channel Dimension)** in the **Supplementary Material**, which presents the full results. The specific added texts are as follows:

Addition to Section III-B of the main manuscript:

“We further analyze the sensitivity of SGFNet to the channel dimension C in the Supplementary Material, which shows that SGFNet achieves the optimal balance between classification accuracy and model complexity when C is set to 64.”

New Section S.II in the Supplementary Material:

“S.II Sensitivity of the Channel Dimension. The channel dimension C determines the feature representation capacity of SGFNet, and may therefore affect both classification performance and model complexity. We evaluate SGFNet with $C \in \{32, 64, 128, 256\}$ on both the Augsburg and Houston2018 datasets, and the results are shown in Table S2. As observed, when C increases from 32 to 64, the OA on the Augsburg and Houston2018 datasets increases significantly, reaching its peak at 92.38% and 94.17%, respectively. This indicates that an appropriate channel dimension effectively enhances the network’s multi-modal feature modeling capability. However, further expanding C to 128 and 256 leads to a continuous performance degradation. This is primarily because an excessively large channel capacity introduces redundant parameters, which easily triggers overfitting on remote sensing datasets with limited training samples. Concurrently, the model complexity scales up rapidly, with the parameter size growing from 0.36M to 16.52M. Based on this analysis, we set $C = 64$ in our experiments, as it achieves the best trade-off between classification accuracy and model complexity.”

We hope this sensitivity analysis of both the dynamic kernel size and the channel dimension effectively addresses the reviewer’s concern. All corresponding changes have been highlighted in red in the updated manuscript and supplementary material for your review. Thank you again for guiding us to improve our manuscript.

TABLE S2 OA (%) and Params (M) of SGFNet under different channel dimensions C .

C	32	64	128	256
Augsburg OA (%)	89.52	92.38	91.14	87.65
Houston2018 OA (%)	91.06	94.17	92.83	89.42
Params (M)	0.36	1.21	4.38	16.52

Reviewer #3, Comment #3

Although SGFNet improves the overall performance, the classification accuracy of difficult categories remains unsatisfactory. The authors should discuss why these classes are still poorly classified.

Response:

We sincerely thank the reviewer for this important comment. We fully agree that a deeper discussion of why certain difficult categories remain poorly classified is necessary for a complete and transparent evaluation of SGFNet. This concern is closely related to comments raised by Reviewer #1, Comment #6 and Reviewer #2, Comment #1, and we have provided a unified and detailed discussion addressing all three comments together.

Specifically, we would like to clarify the following:

(1) Regarding the underlying cause of class imbalance:

As discussed in detail in our response to Comment 1 of Reviewer 2, the underperformance on difficult categories (e.g., Commercial area and Industrial area on the Augsburg dataset) primarily stems from the intrinsic, naturally imbalanced land-cover distribution of the benchmark datasets: these classes correspond to sparse, scattered regions with comparatively few labeled training samples, which limits the supervisory signal available for learning discriminative representations.

(2) Regarding the compounding effect of spectral ambiguity:

Beyond sample scarcity, we further observe that the most difficult categories also exhibit strong spectral and structural similarity to neighboring majority classes. For instance, Commercial area shares overlapping reflectance and geometric characteristics with the neighboring Residential and Industrial area classes, since all correspond to man-made

built-up surfaces. Similarly, on the Houston2018 dataset, the relatively lower accuracy on Roads and Sidewalks reflects the intrinsic spectral similarity between paved urban surfaces, where both the HSI reflectance and the SAR/LiDAR structural cues are weakly discriminative at shared object boundaries. This compounding effect of limited supervision and inherent spectral or structural ambiguity, rather than a specific limitation of the proposed SMCB or FMFB modules, is the primary cause of the remaining classification difficulty on these categories. A detailed class-by-class discussion is provided in the Failure Case Analysis paragraph in Section III-C, added in response to Comment 6 of Reviewer 1.

(3) Regarding preliminary mitigation attempts and future work:

As discussed in our response to Comment 6 of Reviewer 1 and Comment 1 of Reviewer 2, we conducted a preliminary exploration using focal loss and minority-class oversampling, observing a modest improvement on the affected classes. This suggests that while class imbalance is a contributing factor, it is not the sole cause, and the remaining spectral/structural ambiguity requires further investigation, which we have identified as future work in the Conclusion of the revised manuscript.

We note that the corresponding textual additions to **Section III-C (Failure Case Analysis)** and **Section IV (Conclusion)** have already been detailed in our responses to Comment 6 of Reviewer 1 and Comment 1 of Reviewer 2, and are not repeated here to avoid redundancy.

We hope this discussion effectively addresses the reviewer’s concern regarding the remaining classification difficulty on certain categories. All corresponding changes have been highlighted in red in the updated manuscript for your review. Thank you again for guiding us to improve our manuscript.

Reviewer #3, Comment #4

The classification maps of Augsburg are not provided. Please add more visualized classification results.

Response:

We sincerely thank the reviewer for this helpful suggestion. We fully agree that providing

the classification maps on the Augsburg dataset, in addition to the quantitative results in Table I, would offer a more intuitive and complete visual evaluation of SGFNet. We have generated classification maps for all compared methods on the Augsburg dataset. As this is a Letter with a strict page limit, and given that a full-page classification map figure for a second dataset would not fit within the current layout, we have placed this figure in the **Supplementary Material** accompanying the revised manuscript, with a pointer added to the main text.

Specifically, we would like to clarify the following:

(1) Regarding the added visualization:

We have generated classification maps for all compared methods, including S²ENet, DFINet, ExViT, DFFNet, MSFM, GCCQTNNet (the additional baseline introduced in response to Comment 5 of Reviewer 1), and the proposed SGFNet, on the Augsburg dataset, together with the ground truth. This figure follows the same visualization format as Fig. 4, which presents the classification maps on the Houston2018 dataset.

(2) Regarding the observed visual patterns:

Consistent with the quantitative results in Table I, the classification maps show that SGFNet produces visually more coherent and less noisy predictions than the compared methods in the Allotment and Water regions, where SGFNet achieves the best quantitative accuracy, with fewer scattered misclassified pixels and boundaries that align more closely with the ground truth. In the Industrial area, SGFNet’s predictions remain largely consistent with the ground truth, though a modest amount of confusion with neighboring Residential area is still visible, consistent with GCCQTNNet achieving a marginally higher quantitative accuracy on this class. For the Commercial area class, all compared methods, including SGFNet, exhibit pronounced visual confusion with the surrounding Residential and Industrial area regions, with very few pixels correctly classified, consistent with the low quantitative accuracy reported in Table I and the discussion in the Failure Case Analysis.

We have added this new figure as **Fig. S1** in a new **Section S.III (Classification Maps on the Augsburg Dataset)** of the **Supplementary Material**, and added a brief pointer sentence at the end of the Augsburg results discussion in **Section III-C** of the main manuscript. The specific added texts are as follows:

Addition to Section III-C of the main manuscript:

“The corresponding classification maps on the Augsburg dataset are provided in the Supplementary Material due to the page limit of this Letter.”

New Section S.III in the Supplementary Material:

“**S.III Classification Maps on the Augsburg Dataset.** Fig. S1 shows the classification maps of all compared methods on the Augsburg dataset.”

We hope this additional visualization, presented in the Supplementary Material due to the page limit of this Letter, effectively addresses the reviewer’s concern and provides a more complete evaluation of the classification performance on the Augsburg dataset. All corresponding changes have been highlighted in red in the updated manuscript and supplementary material for your review. Thank you again for guiding us to improve our manuscript.

Reviewer #3, Comment #5

The relationship between DCT-domain modulation and spatial misregistration robustness is only qualitatively described. Please provide more evidence or discussions.

Response:

We sincerely thank the reviewer for this important comment. We fully agree that the relationship between DCT-domain modulation and robustness to spatial misregistration required more rigorous evidence beyond a purely qualitative description. This concern is closely related to comments raised by Reviewer 1 (Comment 2) and Reviewer 2 (Comment 2), and we have addressed it with both a more rigorous theoretical discussion and a new controlled experiment providing direct quantitative evidence.

Specifically, we would like to clarify the following:

(1) Regarding the theoretical evidence:

We have grounded our claim in the shift theorem of the discrete transform: a spatial translation of an image alters only the phase component of its frequency representation, while the magnitude/energy distribution across frequency bands remains largely unchanged. Since slight cross-modal misregistration in multi-source remote sensing data

typically manifests as small spatial translations or local geometric distortions, frequency-domain operations that primarily act on the energy distribution, as performed by FMFB, are inherently less sensitive to such shifts than pixel-wise spatial-domain fusion. In addition, the DCT concentrates most of the signal energy into low-frequency components, where the relative effect of small spatial misregistration is proportionally weaker than in high-frequency, edge-sensitive regions. This theoretical basis has been added to Section II-C of the revised manuscript, as detailed in our response to Comment 2 of Reviewer 1.

(2) Regarding the quantitative evidence:

To move beyond a purely qualitative description, we conducted a controlled experiment on the Houston2018 dataset by artificially applying spatial shifts of $\{1, 2, 3, 5\}$ pixels along both the x - and y -directions to the LiDAR branch input, and compared the OA degradation of SGFNet against two representative state-of-the-art methods, DFFNet and MSFMamba, under identical shift conditions. As shown in Table V, at the maximum tested shift of 5 pixels, SGFNet degrades by only 1.40% (from 94.17% to 92.77%), compared to a 2.03% drop for MSFMamba and a 2.60% drop for DFFNet, with the performance gap widening consistently as the shift magnitude increases across all tested levels. The results provide direct quantitative evidence that SGFNet, empowered by DCT-domain modulation in FMFB, exhibits substantially smaller performance degradation as the shift magnitude increases compared to the spatial-domain fusion strategies adopted by DFFNet and MSFMamba, thereby substantiating the claimed relationship between DCT-domain modulation and robustness to spatial misregistration with concrete empirical results rather than qualitative argument alone.

We note that the corresponding textual and experimental additions to **Section II-C** and **Section III-E (Robustness to Spatial Misalignment)** have already been detailed in our responses to Comment 2 of Reviewer 1 and Comment 2 of Reviewer 2, and are not repeated here to avoid redundancy.

We hope this combination of theoretical grounding and quantitative evidence effectively addresses the reviewer’s concern regarding the relationship between DCT-domain modulation and robustness to spatial misregistration. All corresponding changes have been highlighted in red in the updated manuscript for your review. Thank you again for guiding us to improve our manuscript.

In Fig. 3, the role of the frequency-domain difference feature should be further explained. It is unclear whether the difference operation may suppress useful modality-specific information.

Response:

We sincerely thank the reviewer for this insightful comment. We fully agree that the role of the frequency-domain difference feature $\hat{\mathbf{F}}_d$ in Fig. 3 was not sufficiently explained, and that the concern regarding potential suppression of modality-specific information is a valid and important one. We have clarified the design rationale and, importantly, would like to highlight that the architecture of FMFB is specifically designed to avoid this issue.

Specifically, we would like to clarify the following:

(1) Regarding the role of the frequency-domain difference feature:

We would like to emphasize that the difference feature $\hat{\mathbf{F}}_d = \hat{\mathbf{F}}_h - \hat{\mathbf{F}}_x$ (Eq. 9) is used *exclusively* to generate the query and key features $\mathbf{Q}_d = f_q(\hat{\mathbf{F}}_d)$ and $\mathbf{K}_d = f_k(\hat{\mathbf{F}}_d)$ (as shown in Fig. 3), which serve only to compute the cross-modal attention weights in Eqs. (11)–(12). Critically, the value features $\mathbf{V}_h$ and $\mathbf{V}_x$ that carry the actual modality-specific content are separately projected *directly* from the original DCT-transformed features $\hat{\mathbf{F}}_h$ and $\hat{\mathbf{F}}_x$ (Eq. 10), *without* passing through the difference operation. Therefore, the difference feature does not participate in the content pathway that is ultimately propagated and fused; it only modulates *how much* attention is allocated to different frequency components, not *what* content is aggregated.

(2) Regarding why the difference operation does not suppress modality-specific information:

Physically, the frequency-domain difference $\hat{\mathbf{F}}_d$ highlights the frequency components where the two modalities disagree, which typically correspond to regions affected by slight spatial misalignment or genuinely complementary structural information (e.g., edges present in LiDAR but not HSI). Using this discrepancy signal to guide the attention weights allows the network to adaptively emphasize frequency components that require cross-modal reconciliation, while the underlying modality-specific value features $\mathbf{V}_h$ and $\mathbf{V}_x$ remain fully intact and are preserved separately for each modality throughout the attention and

reconstruction process (Eqs. 11–14). Consequently, no modality-specific information is discarded or suppressed; the difference operation only serves as a guidance signal for adaptive cross-modal weighting, not as a transformation applied to the propagated content itself.

We have added the following clarification to **Section II-C (Frequency Modulated Fusion Block)** of the revised manuscript. The specific added text is as follows:

“We emphasize that the frequency-domain difference feature $\hat{\mathbf{F}}_d$ is used exclusively to generate the query and key features that guide the cross-modal attention weights in Eqs. (11)–(12), while the value features $\mathbf{V}_h$ and $\mathbf{V}_x$, which carry the actual modality-specific content, are separately projected directly from the original DCT-transformed features $\hat{\mathbf{F}}_h$ and $\hat{\mathbf{F}}_x$ without passing through the difference operation. As such, $\hat{\mathbf{F}}_d$ only modulates how attention is allocated across frequency components based on cross-modal discrepancy, rather than transforming the content that is ultimately propagated and fused. This design ensures that no modality-specific information is discarded or suppressed by the difference operation.”

We hope this clarification effectively addresses the reviewer’s concern regarding the role of the frequency-domain difference feature and its potential impact on modality-specific information. All corresponding changes have been highlighted in red in the updated manuscript for your review. Thank you again for guiding us to improve our manuscript.

Thanking you again for your precious time and valuable efforts spent in reviewing our manuscript. We greatly appreciate your expertise and the dedicated efforts you have made to assist us in enhancing the quality of this manuscript. We hope that we have clarified all the issues raised by you, and our revised manuscript will again meet your expectations. We eagerly anticipate any additional insights or suggestions you might offer.

Best wishes and thanks again.

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